

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE**

3445UA0-1



APPLIED SCIENCE (Double Award)
UNIT 1: Energy, Resources and the Environment

HIGHER TIER

WEDNESDAY, 12 JUNE 2019 – MORNING

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	11	
2.	8	
3.	11	
4.	11	
5.	6	
6.	12	
7.	16	
Total	75	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

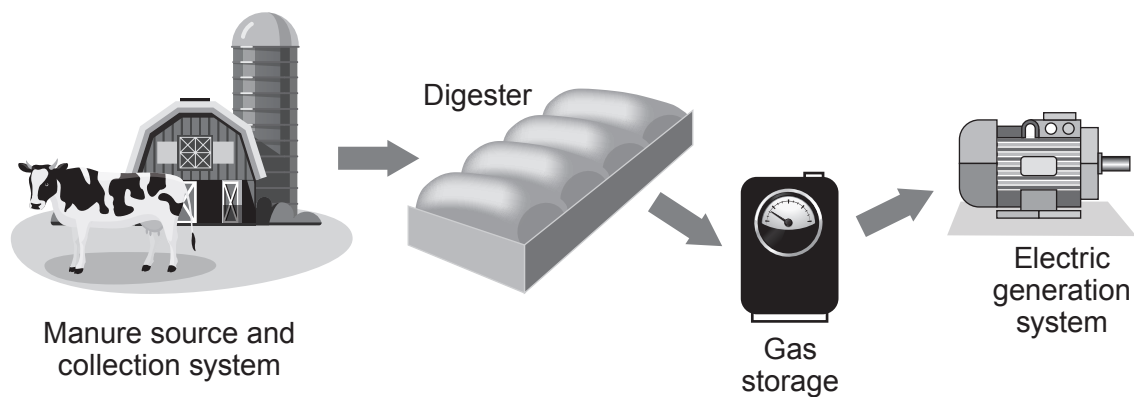
A periodic table is printed on page 16.

Answer all questions.

1. A farmer has installed a biogas generator to save money. The farmer's monthly electricity bill before installation was £3 000. The farmer spent £120 000 to buy and install the biogas generator. After installation the monthly electricity bill was expected to reduce to £600.

The biogas generator works by using the animal waste produced on the farm. This waste is digested by bacteria and the product, methane gas, is used to generate electricity. Methane is a greenhouse gas and produces carbon dioxide and water when it is burned

The animal waste used in the digester is a renewable energy source.



- (a) The biogas generator can provide some of the farm's electricity.

Use the information above to calculate the expected payback time for the biogas generator. [3]

Payback time = months

- (b) The farmer considered installing wind turbines to generate electricity before investing in the biogas generator.

Give **two** disadvantages of using wind power rather than the biogas generator. [2]

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- (c) The biogas generator needs to produce 160 000 kWh of electricity per year to power the farm. A typical cow produces 10 tonnes of waste per year of which the farmer is only able to collect 3 tonnes. For every tonne of waste collected 480 kWh of heat energy is produced, which generates 160 kWh of electricity.

- (i) Use the equation:

$$\% \text{ Efficiency} = \frac{\text{energy usefully transferred}}{\text{total energy supplied}} \times 100$$

to calculate the efficiency of this biogas generator

[2]

% Efficiency =

- (ii) The farmer owns 120 cows. He thinks that he will be able to collect enough waste to power his farm for a year. [4]

Explain whether you agree with the farmer. Show your workings as part of your answer.

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2. In a diffusion experiment, gelatin was mixed with a pink pH indicator and allowed to set. The gelatin was then cut into different sized cubes. The cubes were placed in a beaker containing dilute acid. As the acid diffused through the gelatin the cubes became colourless. The time taken to change from pink to colourless was recorded.

The surface area and volume of the cubes were calculated using the following equations:

surface area of cube = length $\times$ width $\times$ number of sides

volume of cube = length $\times$ width $\times$ height

- (a) The results of the experiment are shown below.

Length of side of gelatin cube (cm)	Time taken to become colourless (s)	Surface area of cube (cm ²)	Volume of cube (cm ³)	Surface area:volume ratio
1	20	6	1	6:1
5	41	150		6:5
7	104		343	6:7
10	610	600	1000	

- (i) **Complete** the table. [3]
- (ii) Sarah suggested that diffusion of the acid into the gelatin is quicker in the smaller cubes because the time taken for them to become colourless is less. Explain whether you agree with Sarah. [2]

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(b) Red blood cells transport oxygen around the body to allow respiration to occur in the tissues. Red blood cells have a volume of 90 units and a surface area of 136 units. Explain how the surface area:volume ratio of red blood cells make them suitable for their function. [3]

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3. In 1912, Alfred Wegener suggested that all of the continents on Earth were once joined together in one supercontinent called Pangaea. These continents have since drifted apart. Wegener suggested a hypothesis called ‘Continental Drift’. In the 1960’s scientists further developed Wegener’s hypothesis using a new theory called ‘Plate Tectonics’.

(a) (i) State **three** pieces of evidence that Wegener used to support his hypothesis of ‘Continental Drift’. [3]

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(ii) Explain the theory of ‘Plate Tectonics’. [3]

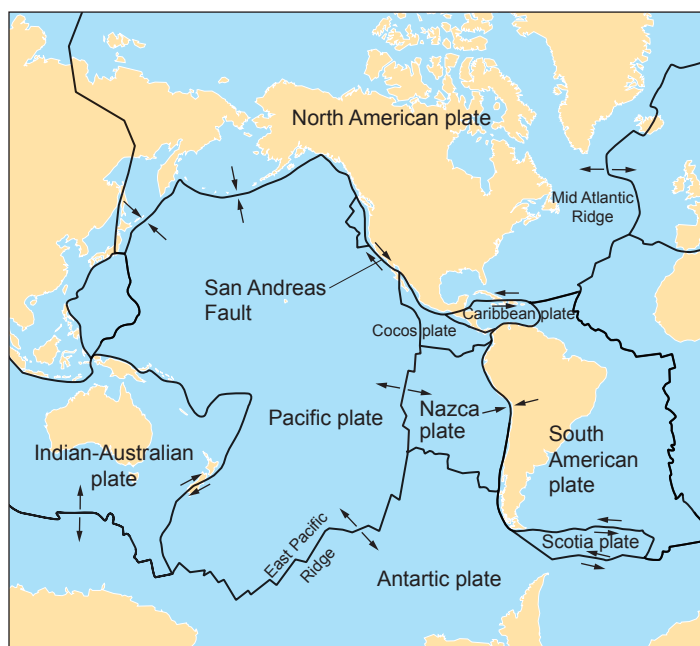
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- (b) The map below shows some of the tectonic plates on Earth and the arrows show the direction of their movement.



Tectonic plates move relative to each other and are monitored closely by geologists using a seismometer. The Pacific and North American plate are very closely monitored by the US government as large populations of people can be affected, especially in the state of California. The boundary between the Pacific and North American tectonic plates is described as a conservative boundary where the plates slide past each other causing powerful earthquakes. The San Andreas Fault lies on this boundary.

Tectonic plate boundaries can also be described as constructive and destructive.

- (i) **Label** on the map **one** constructive (C) and **one** destructive (D) tectonic plate boundary. [1]

- (ii) Describe these **two** types of boundary and the **consequences** of their movement.

Destructive [2]

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Constructive [2]

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4. Port Talbot steel works in South Wales is one of the largest in the UK. It produces 5 million tonnes of steel every year.

The materials required for the production of one tonne of iron are shown in the table below. The melting point of iron is 1535 °C.

- (a) Describe the purpose of the following raw materials in the blast furnace. [2]

Haematite

Limestone

- (b) The blast furnace relies on the processes of reduction and oxidation. In one stage carbon monoxide is formed from coke.

- (i) **Complete** the balanced symbol equation for the extraction of iron in the blast furnace. [3]



- (ii) State the substances that are reduced or oxidised in the reactions that must take place to produce iron in the blast furnace. [2]

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- (c) The materials required for the production of one tonne of iron are shown in the table below. The melting point of iron is 1535 °C.

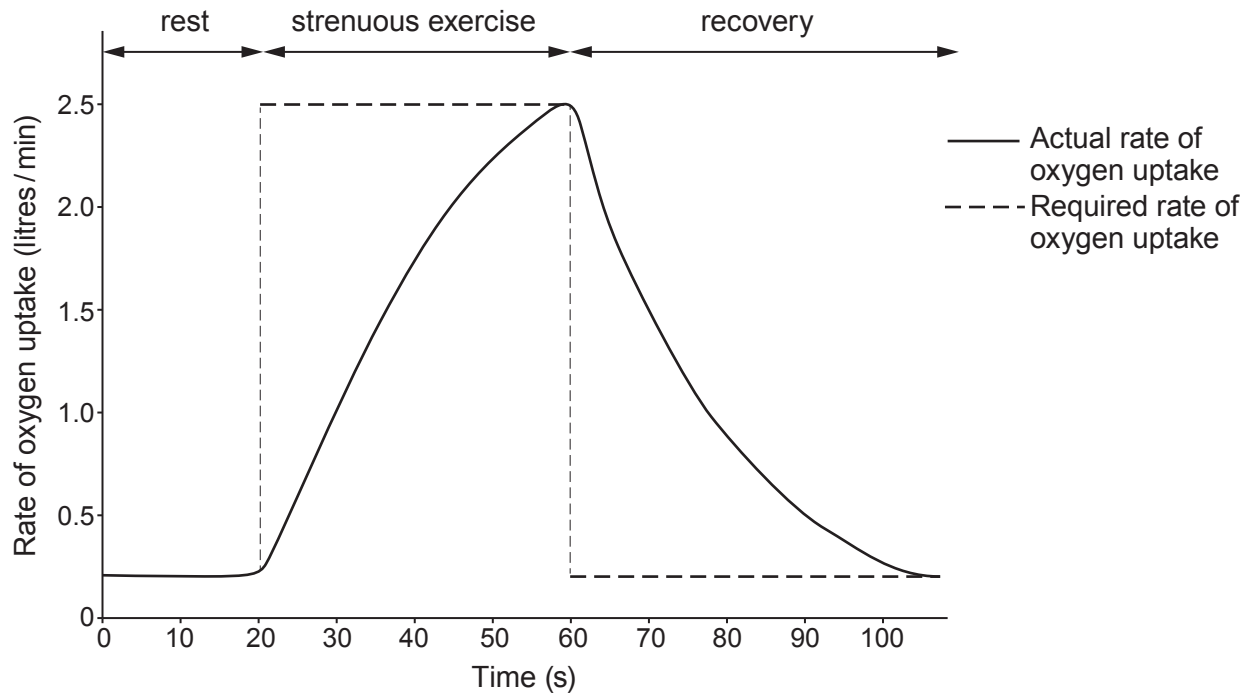
Raw material	Mass of raw material (tonnes)	Cost per tonne of raw material (£)
haematite (iron ore)	1.75	60.50
coke	0.25	120.90
limestone	0.25	80.00
air	4.00	2.25

Calculate the **monthly** raw material cost, for the Port Talbot steel works. Give your answer to two significant figures. [4]

= £ million / month

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5. Respiratory technicians are researching the effect of strenuous exercise on an athlete. They measure how the rate of oxygen uptake can change in an athlete. They do this before, during and after strenuous exercise. This is shown in the graph below.



Use your knowledge and the information in the graph to explain how the type of respiration changes during the 100 seconds shown. [6 QER]

6. Digestion of food is required to provide nutrition for the human body.

(a) (i) Explain how digestion provides nutrition.

[2]

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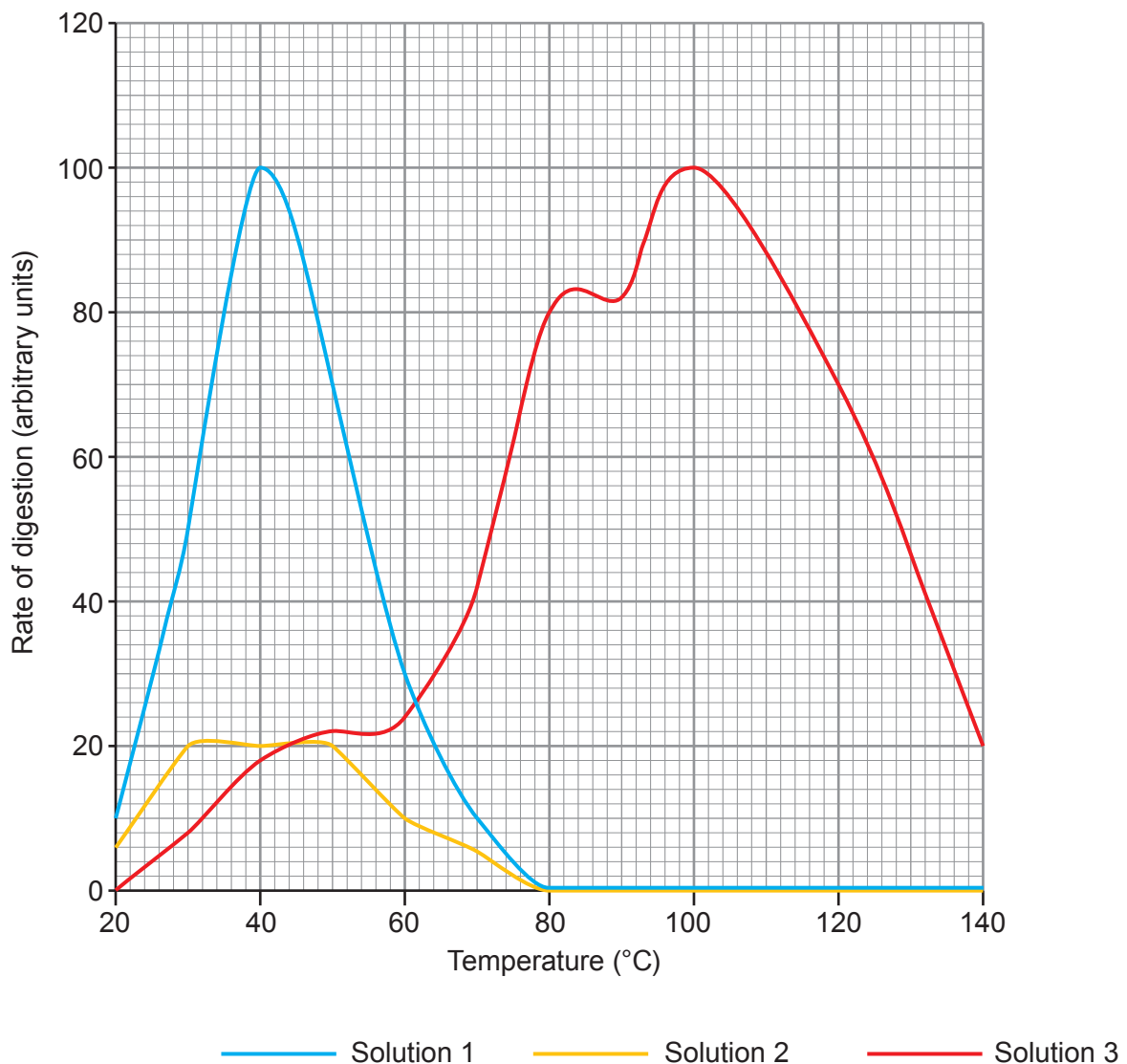
(ii) Describe the process of peristalsis in the digestive system.

[2]

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(b) A group of students were given two solutions of a human enzyme (**solution 1 and solution 2**) and one solution of bacterial enzyme (**solution 3**). All solutions were the same concentration and volume. They measured the rate of digestion using each enzyme solution. A computer calculated the rate of digestion and the results are shown below.



Use the data in the graph and your knowledge to answer the following questions.

- (i) Using the results for **solution 1**, describe and explain the effect of temperature on the rate of digestion between 20 and 60 °C. [4]

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- (ii) Suggest **one** reason why the results shown for **solution 2** are different to **solution 1**. [1]

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- (iii) State a suitable control in this experiment. [1]

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- (iv) **Solution 3** contains an enzyme from a species of bacteria that is found in a hot volcanic region.

Describe the differences between the action of this bacterial enzyme and the enzyme found in humans. [2]

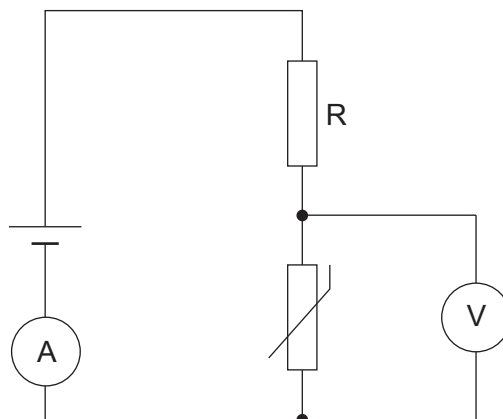
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7. Students were investigating the properties of circuits containing LDRs and thermistors.

(a) The students set up the following circuit.



The temperature of the thermistor was increased.

- (i) On the axes below, sketch how you would expect the resistance of the thermistor to change as the temperature increased. [1]

Resistance
(Ω)



Temperature ($^{\circ}\text{C}$)

(ii) Explain how increasing the temperature of the thermistor affects the readings on the ammeter and voltmeter. [5]

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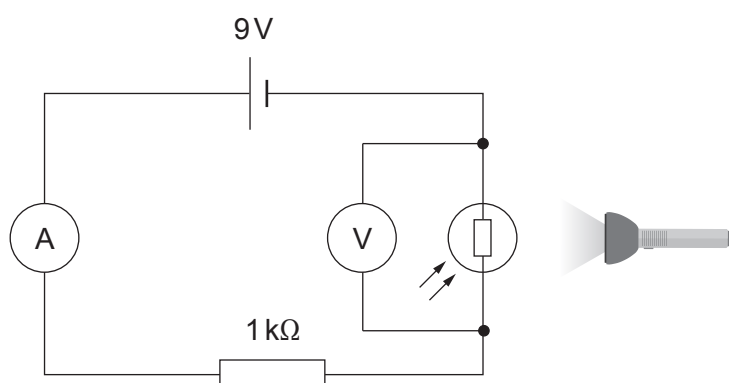
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(b) The students then investigated an LDR with a torch and the following circuit.



(i) State the independent variable in this experiment.

[1]

(ii) State the dependent variable in this experiment.

[1]

(iii) Use the equations:

$$R = R_1 + R_2$$

and

$$V = IR$$

to calculate the readings on the ammeter and the voltmeter in the circuit when the resistance of the LDR is 2.5 kΩ.

[5]

Ammeter: A

Voltmeter: V

- (iv) Describe how you would modify this investigation to determine the I-V characteristics of the LDR. [3]

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END OF PAPER

THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																								4 He Helium 2													
		7 Li Lithium 3		9 Be Beryllium 4																11 B Boron 5		12 C Carbon 6		14 N Nitrogen 7		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10							
		23 Na Sodium 11		24 Mg Magnesium 12																27 Al Aluminium 13		28 Si Silicon 14		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18							
		39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
		86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		128 Te Tellurium 52		127 I Iodine 53		131 Xe Xenon 54	
		133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
		223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																															

Key

1

H

Hydrogen

1

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number